

Technical Document #618: Database Systems - Comprehensive Overview

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Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Sharding distributes data across multiple database servers.
- Database caching stores frequently accessed data in memory.
- Database replication copies data across multiple servers.